

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279010

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

TANAKA; Kenichi et al.

INFORMATION PROCESSING SYSTEM AND COMPUTER SYSTEM IMPLEMENTED METHOD OF PROCESSING INFORMATION

Abstract

An information processing system includes an image obtaining circuit and a display control circuit. The image obtaining circuit is configured to obtain observation images of a first keyboard of a first keyboard instrument. The display control circuit is configured to display, on a display device, the observation images and reference images. The reference images include moving images of at least one hand and one or more fingers of a reference performer who is playing a second keyboard of a second keyboard instrument. The at least one hand and the one or more fingers of the reference performer are displayed overlapping the first keyboard included in the observation images.

Inventors: TANAKA; Kenichi (Hamamatsu-shi, JP), ISOZAKI; Yoshimasa (Hamamatsu-shi, JP), OHKAWA; Haruki (Hamamatsu-shi, JP)

Applicant: YAMAHA CORPORATION (Hamamatsu-shi, JP)

Family ID: 88239642

Appl. No.: 19/202291

Filed: May 08, 2025

Foreign Application Priority Data

JP	2022-063427	Apr. 06, 2022
----	-------------	---------------

Related U.S. Application Data

parent US continuation 18187336 20230321 parent-grant-document US 12322300 child US 19202291

Publication Classification

Int. Cl.: **G09B5/02** (20060101); **G09B15/00** (20060101); **G10G1/02** (20060101); **G10H1/00** (20060101)

U.S. Cl.:

CPC **G09B5/02** (20130101); **G09B15/00** (20130101); **G10G1/02** (20130101); **G10H1/0008** (20130101); **G10H2210/091** (20130101); **G10H2210/385** (20130101); **G10H2220/091** (20130101); **G10H2220/455** (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2022-063427, filed Apr. 6, 2022. The contents of this application are incorporated herein by reference in their entirety.

BACKGROUND

Field

[0002] The present disclosure relates to an information processing system and a computer system implemented method of processing information.

Background Art

[0003] WO/2021/157691 discloses a technique of assisting a user to play a musical instrument such as a keyboard instrument. Specifically, WO/2021/157691 discloses: analyzing an image of hands and fingers of the user who is playing a keyboard instrument; and generating time-series information estimating postures of the hands and fingers.

[0004] When a person practices playing a keyboard instrument, a preferable situation is that a trainer is present near the person. It is, however, difficult for a trainer to be continually present near the person in actual situations. While the foregoing description is under the assumption that a person practices playing a keyboard instrument, similar issues are seen in four or more handed performance of musical instruments such as a keyboard instrument by a plurality of users. The present disclosure has been made in view of the above-described and other problems, and has an object to provide a technique that enables a performer to play a keyboard instrument while looking at performance of another performer.

SUMMARY

[0005] One aspect is an information processing system that includes an image obtaining circuit and a display control circuit. The image obtaining circuit is configured to obtain observation images of a first keyboard of a first keyboard instrument. The display control circuit is configured to display, on a display device, the observation images and reference images. The reference images include moving images of at least one hand and one or more fingers of a reference performer who is playing a second keyboard of a second keyboard instrument. The at least one hand and the one or more fingers of the reference performer are displayed overlapping the first keyboard included in the observation images.

[0006] Another aspect is a computer system implemented method of processing information. The method includes obtaining observation images of a first keyboard of a first keyboard instrument. The method also includes displaying, on a display device, the observation images and reference images, the reference images including a moving image of at least one hand and one or more fingers of a reference performer who is playing a second keyboard of a second keyboard

instrument, such that the at least one hand and the one or more fingers of the reference performer are displayed overlapping the first keyboard included in the observation images.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] A more complete appreciation of the present disclosure and many of the attendant advantages thereof will be readily obtained as the same becomes better understood by reference to the following detailed description when considered in connection with the following figures.

[0008] FIG. 1 is a block diagram of an example configuration of a musical performance system.

[0009] FIG. 2 is a block diagram of an example configuration of an image system.

[0010] FIG. 3 illustrates how an information processing system operates.

[0011] FIG. 4 is a block diagram of an example configuration of the information processing system.

[0012] FIG. 5 is a block diagram of an example functional configuration of the information processing system.

[0013] FIG. 6 is a flowchart of image generation processing.

[0014] FIG. 7 is a flowchart of control processing.

[0015] FIG. 8 schematically illustrates an observation image according to a second embodiment.

[0016] FIG. 9 is a flowchart of the image generation processing according to a fourth embodiment.

[0017] FIG. 10 is a flowchart of the image generation processing according to a fifth embodiment.

[0018] FIG. 11 is a block diagram of an example configuration of the musical performance system according to a sixth embodiment.

[0019] FIG. 12 is a block diagram of an example functional configuration of the information processing system according to the sixth embodiment.

DESCRIPTION OF THE EMBODIMENTS

[0020] The present development is applicable to an information processing system, a computer system implemented method of processing information, and a non-transitory computer-readable recording medium.

A: First Embodiment

[0021] FIG. 1 is a block diagram of an example configuration of a musical performance system **100** according to the first embodiment. The musical performance system **100** is a computer system that assists a user U in playing a keyboard instrument **10**. The user U practices playing the keyboard instrument **10** using the musical performance system **100**. The musical performance system **100** includes the keyboard instrument **10**, a video system **20**, and an information processing system **30**. Each of the keyboard instrument **10** and the video system **20** is communicable with the information processing system **30** via a wire or wirelessly.

[0022] The keyboard instrument **10** is an electronic instrument disposed in a real space where the user U exists. The keyboard instrument **10** includes a keyboard Bx and a housing **11**. The keyboard Bx includes a plurality of keys Kx (white keys and black keys). Each of the plurality of keys Kx corresponds to a different sound pitch. The user U operates the keys Kx of the keyboard instrument **10** sequentially to perform a particular musical piece (hereinafter referred to as “target musical piece”). While the keyboard instrument **10** is being played by the user U, the keyboard instrument **10** supplies performance data P to the information processing system **30**. The performance data P indicates the performance by the user U. The performance data P is data indicating musical notes played by the user U. Every time the user U operates the keyboard Bx (for example, every time a key is pressed), performance data P is output from the keyboard instrument **10**. For example, the performance data P specifies: a sound pitch corresponding to a key Kx operated by the user U; and the strength with which the key was pressed. An example of the performance data P is event data,

which is compliant with MIDI (Musical Instrument Digital Interface) standard.

[0023] The video system **20** is a video device (HMD, Head Mounted Display) mounted the head of the user U. An example of the video system **20** is a HMD goggle or HMD glasses.

[0024] FIG. **2** is a block diagram of an example configuration of the video system **20**. The video system **20** includes an image pick-up device **21** and a display device **22**. That is, the image pick-up device **21** and the display device **22** are mounted on the head of the user U. With this configuration, the image pick-up device **21** and the display device **22** are variable in position and direction depending on the movement of the head of the user U.

[0025] The image pick-up device **21** picks up an image of a predetermined range in a real space. Specifically, the image pick-up device **21** picks up an image of a predetermined range at a front part of the head of the user U (that is, the direction of line of sight). The user U playing the keyboard instrument **10** looks at the hands and fingers, Hx, of the user U and the keyboard Bx. As exemplified in FIG. **3**, the image pick-up device **21** picks up a moving image X (hereinafter referred to as “observation image”). The observation image X includes an image of the right and left hands and fingers Hx of the user U and an image of the keyboard Bx of the keyboard instrument **10**. The range of the image picked up by the image pick-up device **21** is variable based on the movement of the head of the user U. While the user U is playing the keyboard instrument **10**, the image pick-up device **21** generates observation image data Dx. The observation image data Dx indicates the observation image X. The image pick-up device **21** includes, for example: an optical system such as a photographic lens; an image pick-up element that receives incident light from the optical system; and a processing circuit that generates observation image data Dx based on the amount of light received by the image pick-up element. The image pick-up device **21** automatically adjusts the amount of exposure to, for example, ensure clear boundaries between the white keys and the black keys of the keyboard Bx and the hands and fingers Hx of the user U.

[0026] The display device **22** illustrated in FIG. **2** displays various images under the control of the information processing system **30**. Examples of the display device **22** include a liquid-crystal display panel and an organic EL (Electroluminescence) panel. The display device **22** is disposed in front of the eyes of the user U. The display device **22** according to the first embodiment is a non-transmissive display panel, through which light coming from the real space is not transmitted.

[0027] The display device **22** according to the first embodiment displays a guide image Z illustrated in FIG. **3**. The guide image Z is a moving image including the observation image X and a reference image Y. The reference image Y is a three-dimensional moving image including: the hands and fingers, Hy, of a virtual performer who is making a motion of a musical performance (this performer will be hereinafter referred to as “reference performer”); and a keyboard By. The keyboard By is played by the reference performer. The reference performer is a virtual trainer who teaches the user U how to play a musical performance. It is to be noted that the reference performer may be the other performer who plays a duet with the user U. The keyboard By includes a plurality of keys Ky. The plurality of keys Ky are aligned in a lateral direction, similarly to the keyboard Bx in the real space. The reference image Y is a moving image showing how the reference performer is playing a target musical piece by operating the keyboard By using the hands and fingers Hy. That is, the reference image Y shows a fingering model for a target musical piece.

[0028] In the guide image Z, the reference image Y overlaps the keyboard Bx, which is shown in the observation image X. Specifically, the display device **22** displays the guide image Z in a manner in which the hands and fingers Hy of the reference performer and the keyboard By, which are shown in the reference image Y, overlap the keyboard Bx, which is shown in the observation image X. The size and orientation of each of the hands and fingers Hy and the keyboard By, which are shown in the reference image Y, are set based on the size and orientation of the keyboard Bx, which is shown in the observation image X. The reference image Y is enlarged or diminished to, example, make the keys Ky of the keyboard By equivalent in size to the keys Kx of the keyboard Bx, which is shown in the observation image X. Also, the angles of the hands and fingers Hy and

the keyboard **By**, which are shown in the reference image **Y**, are controlled to make the surface of the keyboard **By** parallel to the surface of the keyboard **Bx**, which is shown in the observation image **X**. This configuration enables the user **U** to play the keyboard instrument **10** with a sense of the reference performer's existence in the real space.

[0029] FIG. **4** is a block diagram of an example configuration of the information processing system **30**. The information processing system **30** is a computer system that controls the video system **20**. Specifically, the information processing system **30** displays the guide image **Z** on the display device **22**.

[0030] The information processing system **30** includes a control device **31**, a storage device **32**, an operation device **33**, and a sound emission device **34**. Examples of the information processing system **30** include a portable information device such as a smartphone and a tablet terminal; and a portable or desktop information device such as a personal computer. It is to be noted that the information processing system **30** may be a single device or a combination of a plurality of devices separate from each other. It is also to be noted that the information processing system **30** may be incorporated in the keyboard instrument **10** or the video system **20**.

[0031] The control device **31** is a single processor or a plurality of processors to control the operation of the information processing system **30**. Specifically, the control device **31** includes one or more than one of the following processors: CPU (Central Processing Unit), GPU (Graphics Processing Unit), SPU (Sound Processing Unit), DSP (Digital Signal Processor), FPGA (Field-Programmable Gate Array), and ASIC (Application-Specific Integrated Circuit).

[0032] The storage device **32** is a single memory or a plurality of memories to store a program executed by the control device **31** and various pieces of data used by the control device **31**. Examples of the storage device **32** include a known recording medium such as a semiconductor recording medium and a magnetic recording medium; and a combination of a plurality of kinds of recording mediums. Other examples of the storage device **32** include a portable recording medium attachable and detachable to and from the information processing system **30**; and a cloud storage or a similar recording medium accessible by the control device **31** via a communication network.

[0033] The storage device **32** according to the first embodiment stores musical piece data **M** and reference image data **Dy** of each of a plurality of musical pieces. The musical piece data **M** of each musical piece is data indicating a model or standard performance of the each musical piece. Specifically, the musical piece data **M** of each musical piece is time-series data that specifies the order in time of the musical notes constituting the each musical piece. An example of the musical piece data **M** is a music file compliant with the MIDI standard.

[0034] The reference image data **Dy** of each musical piece is a moving image data indicating the reference image **Y** of the each musical piece. Specifically, the reference image **Y** of each musical piece is a three-dimensional moving image showing how the reference performer is playing the each musical piece. The reference image **Y** of each musical piece can be referred to as a moving image showing a fingering model for the each musical piece.

[0035] The reference image data **Dy** is 3DCG (3-Dimensional Computer Graphics) data of any convenient form. An example of the reference image data **Dy** is point group data using point groups to represent the hands and fingers **Hy** of the reference performer and the keyboard **By**. Another example of the reference image data **Dy** is skeleton data indicating a frame or a joint of each of the hands and fingers **Hy** and the keyboard **By**. For example, the reference image data **Dy** is generated based on images of the hands and fingers **Hy** and the keyboard **By** which images have been picked up from a plurality of directions while a real performer is playing a musical piece. With this configuration, each of the plurality of keys **Ky** of the keyboard **By**, which is shown in the reference image **Y**, is displaced based on a performance played by the reference performer. That is, by watching the reference image **Y**, the user **U** is able to see how the keys **Ky** are displaced, or pressed, by the reference performer.

[0036] The operation device **33** is an input device that receives an operation made by the user **U**.

For example, the user U operates the operation device **33** to select a target musical piece from a plurality of musical pieces. Examples of the operation device **33** include an operator used by the user U; and a touch panel that detects a contact of the user U. It is to be noted that an operation device **33** that is separate from the information processing system **30** may be connected to the information processing system **30** via a wire or wirelessly.

[0037] The sound emission device **34** emits sound under the control of the control device **31**. Examples of the sound emission device **34** include a speaker and a pair of headphones. For example, a sound signal A is supplied to the sound emission device **34**. The sound signal A indicates a waveform of sound corresponding to a motion of a musical performance made by the user U. The sound signal A is converted from a digital signal to an analogue signal by a D/A converter and amplified by an amplifier. The D/A converter and the amplifier, however, are not illustrated for the sake of simplicity. It is to be noted that a sound emission device **34** that is separate from the information processing system **30** may be connected to the information processing system **30** via a wire or wirelessly.

[0038] FIG. **5** is a block diagram of an example functional configuration of the information processing system **30**. The control device **31** executes the program stored in the storage device **32** to implement a plurality of functions to display the guide image Z on the display device **22**. The plurality of functions include an image obtaining circuit **41**, a display control circuit **42**, a reference identification circuit **43**, a sound source processing circuit **44**, and a performance analysis circuit **45**.

[0039] The image obtaining circuit **41** obtains the observation image X from the image pick-up device **21**. Specifically, the image obtaining circuit **41** receives the observation image data Dx, which is generated by the image pick-up device **21**, from the video system **20**.

[0040] The display control circuit **42** displays the guide image Z on the display device **22**. Specifically, in order to display the guide image Z on the display device **22**, the display control circuit **42** outputs guide image data Dz to the display device **22**. The guide image data Dz indicates the guide image Z. In order to generate the guide image Z, the display control circuit **42** synthesizes the observation image X obtained by the image obtaining circuit **41** with the reference image Y of the target musical piece stored in the storage device **32**.

[0041] In generating the guide image Z, the display control circuit **42** synthesizes the reference image Y with the observation image X in a manner in which the hands and fingers Hy of the reference performer and the keyboard By overlap the keyboard Bx, which is shown in the observation image X. The size and orientation of each of the hands and fingers Hy and the keyboard By, which are shown in the reference image Y, are set based on the size and orientation of the keyboard Bx, which is shown in the observation image X. Specifically, the display control circuit **42** enlarges or diminishes the reference image Y to make the keys Ky of the keyboard By equivalent in size to the keys Kx of the keyboard Bx, which is shown in the observation image X. The display control circuit **42** also controls the angles of the hands and fingers Hy and the keyboard By, which are shown in the reference image Y, to make the surface of the keyboard By parallel to the surface of the keyboard Bx, which is shown in the observation image X.

[0042] The reference identification circuit **43** identifies a reference position R in the observation image X. The reference position R serves as a reference for the reference image Y. The reference image Y is located at a position corresponding to the reference position R in the observation image X. In this manner, the guide image Z is generated. In other words, the reference image Y is synthesized with the observation image X in a manner in which the reference image Y has a predetermined position relationship with the reference position R.

[0043] The reference identification circuit **43** analyzes the observation image X to identify the reference position R. Specifically, as exemplified in FIG. **3**, the reference identification circuit **43** detects an end portion **12** of the keyboard Bx, which is shown in the observation image X, and determines the position of the end portion **12** as the reference position R. The end portion **12**

includes: a left end portion **12**, which is on the left side of the keyboard Bx; and a right end portion **12**, which is on the right side of the keyboard Bx. The reference identification circuit **43** detects one or both of the left end portion **12** and the right end portion **12**. Any known object-detection technique may be used to detect the end portion **12** of the keyboard Bx. For example, object detection processing that uses a trained model such as a deep neural network may be used to detect the end portion **12** (that is, identify the reference position R) by the reference identification circuit **43**. It is to be noted that the end portion **12** of the keyboard Bx is clearly distinguished from the left and right ends of the housing **11** of the keyboard instrument **10** (the left and right ends are respectively located on the left side and right side of the end portion **12** of the keyboard Bx) in terms of optical properties such as color and material. Therefore, the end portion **12** of the keyboard Bx can be accurately and easily detected based on the observation image X.

[0044] The sound source processing circuit **44** illustrated in FIG. 5 generates a sound signal A that is based on an operation that the user U has made on the keyboard Bx. As described above, the performance data P specifies a sound pitch corresponding to the key Kx operated by the user U, and this performance data P is supplied to the information processing system **30** from the keyboard instrument **10**. The sound source processing circuit **44** is a MIDI sound source that generates a sound signal A corresponding to the sound pitch specified by the performance data P. That is, the sound signal A is a signal indicating a sound waveform of a sound pitch corresponding to one or more keys Kx operated by the user U. In the first embodiment, the sound source processing circuit **44** is a software sound source implemented by the control device **31**. Another possible example of the sound source processing circuit **44** is a hardware sound source dedicated to generating the sound signal A.

[0045] The performance analysis circuit **45** analyzes the performance of the user U playing the keyboard instrument **10**. The performance analysis circuit **45** according to the first embodiment uses the performance data P supplied from the keyboard instrument **10** to analyze the performance of the user U. The performance analysis circuit **45** according to the first embodiment identifies a performance point Q1, a performance speed Q2, and an operation position Q3. The performance analysis circuit **45** repeats this analysis at predetermined time intervals.

[0046] The performance point Q1 is a point of time on a time axis of the target musical piece that is being played by the user U. Any known technique may be used to identify the performance point Q1. For example, in identifying the performance point Q1, the performance analysis circuit **45** compares the time order of the pieces of performance data P supplied from the keyboard instrument **10** with the musical piece data M of the target musical piece.

[0047] The performance speed Q2 is a speed (tempo) at which the user U plays the target musical piece. The performance analysis circuit **45** identifies the performance speed Q2 based on a change in time of the performance point Q1. For example, the displacement amount of the performance point Q1 per unit time is calculated as the performance speed Q2.

[0048] The operation position Q3 is a spatial position on the keyboard Bx of the keyboard instrument **10** which spatial position is where the user U is playing. As described above, the performance data P specifies a sound pitch played by the user U. The performance analysis circuit **45** determines the sound pitch specified by the performance data P as the operation position Q3.

[0049] The display control circuit **42** generates the guide image Z using the reference position R identified by the reference identification circuit **43** and using the analysis results (Q1 to Q3) obtained by the performance analysis circuit **45**. FIG. 6 is a flowchart of image generation processing S5. The image generation processing S5 is performed by the display control circuit **42** in generating the guide image Z.

[0050] Upon start of the image generation processing S5, the display control circuit **42** controls the reference image Y to be reproduced based on the performance point Q1 and the performance speed Q2 (S51). From the storage device **32**, the display control circuit **42** sequentially obtains images of parts of the reference image Y of the target musical piece which parts correspond to the

performance point **Q1**. For example, there may be a case where the performance point **Q1** is a point of time at which the reference performer plays a particular musical note of the target musical piece. In this case, the display control circuit **42** obtains an image that is included in the reference image **Y** and that corresponds to the point of time at which the reference performer plays the particular musical note. The display control circuit **42** also controls the reproduction speed of the reference image **Y** based on the performance speed **Q2**. For example, as the performance speed **Q2** is higher, the reproduction speed of the reference image **Y** increases. That is, the display control circuit **42** according to the first embodiment causes the reproduction of the reference image **Y** to proceed along with the user **U**'s performance of the target musical piece. In other words, the display control circuit **42** controls the reference image **Y** to be reproduced so that the user **U**'s performance of the target musical piece is synchronized with the reproduction of the reference image **Y**.

[0051] The display control circuit **42** sets the position of the reference image **Y** relative to the observation image **X** based on the operation position **Q3** and the reference position **R** (**S52**). Specifically, the display control circuit **42** identifies the operation position **Q3** in the keyboard **Bx** based on the reference position **R** (that is, the end portion **12** of the keyboard **Bx**). Then, the display control circuit **42** identifies a position that has a predetermined relationship with the operation position **Q3**, and determines this position as the position of the reference image **Y**. For example, the display control circuit **42** arranges (displays) the reference image **Y** in an area of the keyboard **Bx** which area is higher or lower in sound than a predetermined area of the keyboard, the predetermined area including the operation position **Q3**. An example of the predetermined area including the operation position **Q3** is a range of area centered around the operation position **Q3** and one octave higher or lower than the operation position **Q3**. With this configuration, the reference image **Y** is arranged at a position that is one octave higher or lower than the sound pitch (the operation position **Q3**) at which the user **U** is playing. In other words, the position of the reference image **Y** is set in a manner in which the reference image **Y** does not overlap the user **U**'s hands and fingers **Hx**, which are shown in the observation image **X**. This configuration enables the user **U** to easily look at the reference image **Y** while playing the keyboard instrument **10**.

[0052] The display control circuit **42** synthesizes the reference image **Y** with the observation image **X** to generate the guide image **Z** (**S53**). Specifically, a part of the reference image **Y** which part corresponds to the performance point **Q1** and the performance speed **Q2** is synthesized with the observation image **X** at a position in the observation image **X** which position is set based on the operation position **Q3** and the reference position **R**. The display control circuit **42** controls the size and the angle of the reference image **Y** in a manner in which the reference image **Y** overlaps the keyboard **Bx**, which is shown in the observation image **X**.

[0053] The position or the angle of the keyboard **Bx**, which is shown in the observation image **X**, is variable based on the movement of the head of the user **U**. The reference image **Y** is arranged so as to overlap the keyboard **Bx**, which is shown in the observation image **X**. This configuration ensures that the positions and the angles of the hands and fingers **Hy** of the reference performer and the position and the angle of the keyboard **By** are also variable based on the movement of the head of the user **U** when the hands and fingers **Hy** and the keyboard **By** are displayed on the display device **22**. In other words, by changing the position and the direction of the head of the user **U** in any manner deemed convenient, the user **U** is able to look at the hands and fingers **Hy** of the reference performer and the keyboard **By** from a desired position and a desired angle. As described above, the keys **Ky** of the keyboard **By**, which is shown in the reference image **Y**, are displaced based on the reference performer's motion of a musical performance. Also in the guide image **Z**, which is displayed on the display device **22**, the keys **Ky** of the keyboard **By** are displaced based on motions of the hands and fingers **Hy** associated with the musical performance. This configuration enables the user **U** to visually see the hands and fingers **Hy** of the reference performer with a sense of the reference performer's actual operation of the keyboard **Bx** of the keyboard instrument **10**.

[0054] FIG. **7** is a flowchart of control processing. The control processing is performed by the

control device **31**. The control processing exemplified below is repeated at predetermined time intervals.

[0055] Upon start of the control processing, the control device **31** (the image obtaining circuit **41**) obtains an observation image X (**S1**). The reference identification circuit **43** analyzes the observation image X to identify the reference position R (**S2**). The control device **31** (the sound source processing circuit **44**) generates a sound signal A based on performance data P supplied from the keyboard instrument **10** (**S3**). The control device **31** (the performance analysis circuit **45**) analyzes the performance of the user U playing the keyboard instrument **10** to identify an operation position Q3, a performance point Q1, and a performance speed Q2 (**S4**). The order of the above-described steps (**S1** to **S4**) of the control processing may be changed in any manner deemed convenient.

[0056] The control device **31** (the display control circuit **42**) performs the image generation processing **S5** exemplified in FIG. **6** to generate a guide image Z. As described above, the control device **31** (the display control circuit **42**) synthesizes the observation image X with the reference image Y to generate the guide image Z. The control device **31** (the display control circuit **42**) outputs guide image data Dz, which indicates the guide image Z, to the display device **22** to display the guide image Z on the display device **22** (**S6**).

[0057] As described above, in the first embodiment, the hands and fingers Hy of the reference performer, which is shown in the reference image Y, are displayed in a manner in which the hands and fingers Hy overlap the keyboard Bx, which is shown in the observation image X. This configuration provides the user U with an effective customer experience of playing the keyboard instrument **10** while looking at the reference performer's hands and fingers Hy superimposed on the keyboard Bx of the keyboard instrument **10**. In the first embodiment, the positions and the angles of the image pick-up device **21** and the display device **22** are variable based on the position and the angle of the head of the user U. Along with a motion (for example, movement or rotation) of the head of the user U, the keyboard By is also variable in position and angle in the reference image Y displayed on the display device **22**. This configuration provides the user U with a customer experience of playing the keyboard instrument **10** with a sense of the reference performer's existence in the real space.

[0058] Also in the first embodiment, the reference image Y is displayed at a position in the observation image X which position corresponds to the reference position R on the keyboard instrument **10**. This configuration ensures that an image of the hands and fingers Hy of the reference performer is arranged at a position in the observation image X which position is appropriate relative to the keyboard instrument **10** in the real space.

[0059] In the first embodiment, the reproduction of the reference image Y is controlled based on the performance point Q1 and the performance speed Q2. That is, the reproduction of the reference image Y proceeds along with the user U's performance of a target musical piece. This configuration enables the user U to play the target musical piece with a sense of cooperating with the reference performer on the keyboard instrument **10**.

B: Second Embodiment

[0060] The second embodiment will be described. It is to be noted that like reference numerals designate corresponding or identical elements throughout the above and following embodiments, and these elements will not be elaborated upon here. In the first embodiment, the end portion **12** of the keyboard Bx, which is shown in the observation image X, is identified as the reference position R. The second embodiment and the third embodiment are different from the first embodiment in how to identify the reference position R, and otherwise similar to the first embodiment.

[0061] FIG. **8** schematically illustrates an observation image X according to the second embodiment. In the second embodiment, the keyboard instrument **10** is provided with a mark **13**. In the example illustrated in FIG. **8**, the mark **13** has a star shape. The mark **13** is formed on the exterior of the keyboard instrument **10**. For example, the mark **13** is formed at a center portion in

the lateral direction of the housing **11** of the keyboard instrument **10**. The mark **13** is a graphical object or a symbol that visually shows a trade name of the keyboard instrument **10** or a provider (manufacturer or distributor) of the keyboard instrument **10**. Typically, the mark **13** is a trademark. The mark **13** is formed on the housing **11** by, for example, coating or printing. Another possible example is that a planar member with the mark **13** formed on the surface is fixed to the housing **11** or that a planar member formed in the shape of the mark **13** is fixed to the housing **11**.

[0062] The reference identification circuit **43** analyzes the observation image X to identify the reference position R (S2). The reference identification circuit **43** according to the second embodiment detects the mark **13** on the keyboard instrument **10**, which is shown in the observation image X. Then, the reference identification circuit **43** determines the position of the mark **13** as the reference position R. Any known object-detection processing may be used to detect the mark **13**. For example, object detection processing that uses a trained model such as a deep neural network may be used to detect the mark **13** (that is, identify the reference position R) by the reference identification circuit **43**. Then, the position of the reference image Y relative to the observation image X is set based on the reference position R (S52). This processing is similar to the corresponding processing in the first embodiment.

[0063] The second embodiment provides advantageous effects similar to the advantageous effects provided in the first embodiment. It is to be noted that in many cases, the mark **13** on the keyboard instrument **10** is formed at a particular position on the keyboard instrument **10**. In the second embodiment, the position of the mark **13** on the keyboard instrument **10** is identified as the reference position R. This configuration ensures that the reference position R is identified accurately even if such a situation occurs that the end portion **12** of the keyboard Bx can not be detected. For example, there may be a case where the end portion **12** of the keyboard Bx is located outside the image pick-up range of the image pick-up device **21** (a situation where the end portion **12** is not included in the observation image X). Even in this case, the reference position R can be identified in the second embodiment.

C: Third Embodiment

[0064] In the third embodiment, the reference identification circuit **43** identifies the position of a predetermined, particular key (hereinafter referred to as “particular key”) Kx from among the plurality of keys Kx of the keyboard Bx of the keyboard instrument **10**. Then, the reference identification circuit **43** determines the particular key Kx as the reference position R.

[0065] Before the guide image Z is displayed, the user U operates the particular key Kx of the keyboard Bx. The reference identification circuit **43** analyzes the observation image X to identify the reference position R (S2). The reference identification circuit **43** according to the third embodiment analyzes the observation image X to detect the key Kx operated by the user U in the observation image X (that is, detect the particular key Kx). Then, the reference identification circuit **43** determines the position of the particular key Kx as the reference position R. For example, a key Kx among the plurality of keys Kx shown in the observation image X is regarded as the particular key Kx if there was a change in tone at this key Kx upon operation by the user U. Then, the position of the reference image Y relative to the observation image X is set based on the reference position R (S52). This processing is similar to the corresponding processing in the first embodiment.

[0066] The third embodiment provides advantageous effects similar to the advantageous effects provided in the first embodiment. Also in the third embodiment, the position of the particular key Kx operated by the user U is identified as the reference position R. This configuration ensures that the reference position R is identified accurately even in a situation where the end portion **12** or the mark **13** on the keyboard instrument **10** can not be detected. For example, there may be a case where the end portion **12** and the mark **13** on the keyboard instrument **10** are located outside the image pick-up range of the image pick-up device **21**. Even in this case, the reference position R can be identified in the third embodiment.

D: Fourth Embodiment

[0067] The fourth embodiment is different from the first embodiment in the operation of the performance analysis circuit **45** and the operation of the display control circuit **42**. The performance analysis circuit **45** according to the fourth embodiment generates information (Q1 to Q3) similar to the information (Q1 to Q3) generated in the first embodiment. The performance analysis circuit **45** according to the fourth embodiment also determines whether the user U is playing the keyboard instrument **10**. Specifically, when performance data P has been supplied from the keyboard instrument **10**, the performance analysis circuit **45** determines that the keyboard instrument **10** is being played. When no performance data P has been supplied from the keyboard instrument **10**, the performance analysis circuit **45** determines that the keyboard instrument **10** is not being played. This manner of determination by the performance analysis circuit **45** is not intended in a limiting sense. For example, the performance analysis circuit **45** may analyze the observation image X to determine whether the keyboard instrument **10** is being played or not.

[0068] The display control circuit **42** synthesizes the observation image X with the reference image Y to generate the guide image Z, similarly to the display control circuit **42** according to the first embodiment. In generating the guide image Z, the display control circuit **42** according to the fourth embodiment controls the numerical value of transmission parameter α . The transmission parameter α is associated with the reference image Y. Specifically, the transmission parameter α serves as an indicator (alpha value) of the degree at which an element behind the reference image Y is transmitted through the reference image Y. In the fourth embodiment, the transmission parameter α is the degree of transparency. Specifically, as the numerical value of the transmission parameter α is larger, the degree at which the element behind the reference image Y is transmitted through the reference image Y is higher. For example, when the transmission parameter α is set to 0%, the reference image Y is displayed as a non-transparent image. When the transmission parameter α is set to 50%, the reference image Y is displayed as a semi-transparent image. When the reference image Y is displayed as a semi-transparent image, the user U is able to look at the part of the observation image X which part is located behind the reference image Y.

[0069] The display control circuit **42** controls the numerical value of the transmission parameter α based on whether the keyboard instrument **10** is being played or not. FIG. 9 is a flowchart of the image generation processing S5 according to the fourth embodiment. The processing of controlling the reproduction of the reference image Y (S51) and the processing of controlling the position of the reference image Y (S52) are similar to the respective, corresponding processings performed in the first embodiment.

[0070] The control device **31** (the performance analysis circuit **45**) determines whether the keyboard instrument **10** is being played (S54). When the keyboard instrument **10** is not being played (S54: NO), the control device **31** (the display control circuit **42**) sets the transmission parameter α to a numerical value of α_1 (S55). When the keyboard instrument **10** is being played (S54: YES), the control device **31** (the display control circuit **42**) sets the transmission parameter α to a numerical value of α_2 (S56). Based on the transmission parameter α obtained in the above-described processing, the control device **31** (the display control circuit **42**) synthesizes the reference image Y with the observation image X to generate the guide image Z (S53). The fourth embodiment is similar to the first embodiment other than the image generation processing S5.

[0071] Numerical value α_2 of the transmission parameter α is higher in transmission degree than numerical value α_1 . As described above, the transmission parameter α according to the fourth embodiment is the degree of transparency. Numerical value α_2 is set to a numerical value larger than numerical value α_1 . For example, numerical value α_2 is set to 50% (semi-transparent), and numerical value α_1 is set to 0% (non-transparent). Thus, when the keyboard instrument **10** is not being played, the reference image Y is displayed as a non-transparent image ($\alpha=\alpha_1$). When the keyboard instrument **10** is being played, the reference image Y is displayed as a semi-transparent image ($\alpha=\alpha_2$). That is, when the keyboard instrument **10** is in a state of being played by the user U,

the degree of transmission of the reference image Y is higher than when the keyboard instrument **10** is in a state of not being played by the user U. It is to be noted that numerical value $\alpha 1$ is an example of the “first value”, and numerical value $\alpha 2$ is an example of the “second value”.

[0072] The fourth embodiment provides advantageous effects similar to the advantageous effects provided in the first embodiment. In the fourth embodiment, the numerical value of the transmission parameter α of the reference image Y is controlled. This control ensures such a usage form that the reference image Y is prioritized for attention from the user U, and such a usage form that the hands and fingers Hx of the user U are prioritized for attention from the user U. These usage forms are switchable to and from each other. This switchable configuration is in contrast to a configuration in which the numerical value of the transmission parameter α is fixed.

[0073] For example, in the fourth embodiment, when the user U is playing the keyboard instrument **10**, the transmission parameter α is set to numerical value $\alpha 2$, which is higher in transmission degree than numerical value $\alpha 1$. This configuration enables the user U to prioritize and focus on the hands and fingers Hx of the user U itself. This configuration is in contrast to a configuration in which the transmission parameter α is fixed to numerical value $\alpha 1$ (a configuration in which the reference image Y is continually displayed as a non-transparent image). When the user U is not playing the keyboard instrument **10**, the transmission parameter α is set to numerical value $\alpha 1$, which is lower in transmission degree than numerical value $\alpha 2$. This configuration enables the user U to easily look at the reference image Y. This configuration is in contrast to a configuration in which the transmission parameter α is fixed to numerical value $\alpha 2$ (a configuration in which the reference image Y is continually displayed as a semi-transparent image). It is to be noted that the second embodiment or the third embodiment is applicable to the fourth embodiment.

E: Fifth Embodiment

[0074] The fifth embodiment is similar to the fourth embodiment in that the numerical value of the transmission parameter α , which is applied to the reference image Y, is controlled in a variable manner. In the fourth embodiment, the transmission parameter α is controlled based on whether the keyboard instrument **10** is being played or not. In the fifth embodiment, the transmission parameter α is controlled based on a motion mode. In the fifth embodiment, the transmission parameter α is the degree of transparency, similarly to the transmission parameter α according to the fourth embodiment. Specifically, as the numerical value of the transmission parameter α is larger, the degree at which the element behind the reference image Y is transmitted through the reference image Y is higher.

[0075] The display control circuit **42** sets one of a first motion mode and a second motion mode as the motion mode. The first motion mode is a reference mode in which the reference image Y is prioritized for attention (visual recognition). The second motion mode is a practice mode in which the hands and fingers Hx of the user U are prioritized for attention. For example, the display control circuit **42** sets the motion mode based on an instruction input on the operation device **33** from the user U. Specifically, every time the user U instructs the motion mode to be switched, the display control circuit **42** switches the motion mode between the first motion mode and the second motion mode.

[0076] FIG. **10** is a flowchart of the image generation processing S5 according to the fifth embodiment. The processing of controlling the reproduction of the reference image Y (S51) and the processing of controlling the position of the reference image Y (S52) are similar to the respective, corresponding processings performed in the first embodiment.

[0077] The control device **31** (the display control circuit **42**) determines whether the motion mode is the first motion mode (S57). When the motion mode is the first motion mode (S57: YES), the control device **31** (the display control circuit **42**) sets the transmission parameter α to a numerical value of $\alpha 1$ (S58). When the motion mode is the second motion mode (S57: NO), the control device **31** (the display control circuit **42**) sets the transmission parameter α to a numerical value of $\alpha 2$ (S59). Based on the transmission parameter α obtained in the above-described processing, the

control device **31** (the display control circuit **42**) synthesizes the reference image Y with the observation image X to generate the guide image Z (S53). The fifth embodiment is similar to the first embodiment other than the image generation processing S5.

[0078] Similarly to the fourth embodiment, numerical value α_2 of the transmission parameter α is higher in transmission degree than numerical value α_1 . Specifically, numerical value α_2 is set to a numerical value larger than numerical value α_1 . For example, numerical value α_2 is set to 50% (semi-transparent), and numerical value α_1 is set to 0% (non-transparent). Thus, in the first motion mode, the reference image Y is displayed as a non-transparent image ($\alpha=\alpha_1$). In the second motion mode, the reference image Y is displayed as a semi-transparent image ($\alpha=\alpha_2$). Specifically, in the first motion mode, the degree of transmission of the reference image Y is higher than in the second motion mode. It is to be noted that numerical value α_1 is an example of the “first value”, and numerical value α_2 is an example of the “second value”.

[0079] The fifth embodiment provides advantageous effects similar to the advantageous effects provided in the first embodiment. In the fifth embodiment, the numerical value of the transmission parameter α of the reference image Y is controlled. This control, similarly to the fourth embodiment, ensures such a usage form that the reference image Y is prioritized for attention from the user U, and such a usage form that the hands and fingers Hx of the user U are prioritized for attention from the user U. These usage forms are switchable to and from each other.

[0080] For example, in the fifth embodiment, in the second motion mode, the transmission parameter α is set to numerical value α_2 , which is higher in transmission degree than numerical value α_1 . This configuration enables the user U to prioritize and focus on the hands and fingers Hx of the user U itself. This configuration is in contrast to a configuration in which the transmission parameter α is fixed to numerical value α_1 (a configuration in which the reference image Y is continually displayed as a non-transparent image). In the first motion mode, the transmission parameter α is set to numerical value α_1 , which is lower in transmission degree than numerical value α_2 . This configuration enables the user U to easily look at the reference image Y. This configuration is in contrast to a configuration in which the transmission parameter α is fixed to numerical value α_2 (a configuration in which the reference image Y is continually displayed as a semi-transparent image). It is to be noted that the second embodiment or the third embodiment is applicable to the fifth embodiment.

F: Sixth Embodiment

[0081] FIG. **11** is a block diagram of an example configuration of the musical performance system **100** according to the sixth embodiment. In the musical performance system **100** according to the sixth embodiment, the information processing system **30** includes a communication device **35** and elements that are similar to the elements in the first embodiment (the control device **31**, the storage device **32**, the operation device **33**, and the sound emission device **34**). The communication device **35** communicates with an external apparatus or device via a communication network **200**, which is the Internet or another network. It is to be noted that the communication between the communication device **35** and the communication network **200** may be wired and wireless. It is also to be noted that a communication device **35** that is separate from the information processing system **30** may be connected to the information processing system **30** via a wire or wirelessly.

[0082] As illustrated in FIG. **11**, a recording system **60** is a computer system that records the reference image Y and is used when a trainer T teaches the user U how to play the keyboard instrument **10**. The trainer T corresponds to the reference performer. Specifically, in the sixth embodiment, the reference image Y is a moving image showing how the trainer T plays a keyboard instrument **63**. An example of the keyboard instrument **63** is a natural musical instrument that includes the keyboard By or an electronic instrument that includes the keyboard By. The recording system **60** includes an image pick-up device **61** and a communication device **62**.

[0083] The image pick-up device **61** picks up an image of how the trainer T plays the keyboard instrument **63**. Specifically, the image pick-up device **61** picks up a reference image Y. The

reference image Y includes an image of the right and left hands and fingers Hy of the trainer T and an image of the keyboard By of the keyboard instrument **63**. Examples of the communication device **62** include a portable information device such as a smartphone and a tablet terminal; and a portable or desktop information device such as a personal computer. The image pick-up device **61** may be incorporated in the communication device **62**.

[0084] The communication device **62** sends reference image data Dy to the information processing system **30**. The reference image data Dy indicates the reference image Y. Specifically, the reference image data Dy is a moving image data indicating a motion of a musical performance which motion is made by the trainer T (reference performer). An example of the reference image data Dy is point group data or skeleton data that indicates the hands and fingers Hy of the trainer T and the keyboard By, similarly to the first embodiment. While the trainer T is playing the keyboard instrument **63**, the reference image data Dy is send to the musical performance system **100** from the recording system **60**. The reference image data Dy is received by the communication device **35** of the information processing system **30** via the communication network **200**.

[0085] FIG. **12** is a block diagram of an example functional configuration of the information processing system **30** according to the sixth embodiment. The control device **31** according to the sixth embodiment serves as a reception control circuit **46** and as elements similar to the elements in the first embodiment (the image obtaining circuit **41**, the display control circuit **42**, the reference identification circuit **43**, the sound source processing circuit **44**, and the performance analysis circuit **45**). The reception control circuit **46** receives, at the communication device **35**, the reference image data Dy sent from the recording system **60**.

[0086] The display control circuit **42** according to the first embodiment generates the guide image Z by synthesizing the observation image X with the reference image Y of the reference image data Dy stored in the storage device **32**. The display control circuit **42** according to the sixth embodiment generates the guide image Z by synthesizing the observation image X with the reference image Y of the reference image data Dy obtained by the reception control circuit **46** (S5). Specifically, the display control circuit **42** displays, on the display device **22**, the reference image Y indicated by the reference image data Dy. Thus, in the sixth embodiment, the guide image Z includes the observation image X and the reference image Y. The observation image X shows how the user U plays the keyboard instrument **10**. The reference image Y shows how the trainer T plays the keyboard instrument **63**. While the trainer T is playing the keyboard instrument **63**, the display device **22** displays the guide image Z in a real time manner. The guide image Z includes the reference image Y of the musical performance.

[0087] The sixth embodiment provides advantageous effects similar to the advantageous effects provided in the first embodiment. In the sixth embodiment, while an image of the trainer T's (reference performer's) motion of a musical performance is being picked up, the reference image Y, which shows this motion, can be displayed on the user U's display device **22**. This configuration enables the user U to play the keyboard instrument **10** with a sense of the trainer T's existence near the user U, even though the trainer T is actually remote from the user U.

G: Modifications

[0088] Modifications of the above-described embodiments will be described below. An embodiment(s) arbitrarily selected from the above-described embodiments may be combined with a modification(s) arbitrarily selected from the following modifications insofar no contradiction occurs.

[0089] (1) In the above-described embodiments, the display device **22** is a non-transmissive display panel. The display device **22** may, however, be a transmissive display panel that transmits light coming from the real space. In a modification in which a transmissive display device **22** is used, it is not essential to display the observation image X on the display device **22**. Specifically, the display control circuit **42** displays, as the guide image Z, the reference image Y on the display device **22**. This configuration enables the user U to look at the reference image Y over the keyboard

Bx while looking at the keyboard Bx of the keyboard instrument **10** utilizing transmitted light from the real space. In this modification in which a transmissive display device **22** is used, the observation image X is used to identify the reference position R on the reference image Y, similarly to the above-described embodiments. As seen from the above description, the observation image X may be omitted. The display device **22** displays the reference image Y using augmented reality (AR) or mixed reality (MR) technology.

[0090] (2) In the fourth and fifth embodiments, the transmission parameter α is the degree of transparency. The transmission parameter α may, however, be the degree of non-transparency. In a modification in which the transmission parameter α is the degree of non-transparency, as the numerical value of the transmission parameter α is larger, the degree at which the element behind the reference image Y is transmitted through the reference image Y is lower. In this modification, when the keyboard instrument **10** is not being played, the transmission parameter α is set to a numerical value of α_1 . When the keyboard instrument **10** is being played, the transmission parameter α is set to a numerical value of α_2 . This configuration is similar to the fourth embodiment. Also in this modification, in the first motion mode, the transmission parameter α is set to a numerical value of α_1 . In the second motion mode, the transmission parameter α is set to a numerical value of α_2 . This configuration is similar to the fifth embodiment. It is to be noted, however, that when the transmission parameter α is the degree of non-transparency, numerical value α_2 is smaller than numerical value α_1 . For example, numerical value α_2 is set to 50% (semi-transparent), and numerical value α_1 is set to 100% (non-transparent). As seen from the above description, numerical value α_2 is generally referred to as a numerical value higher in transmission degree than numerical value α_1 , irrespective of whether the transmission parameter α is degree of transparency or the degree of non-transparency. Whether the numerical value of the transmission parameter α is large or small may have any relationship with whether the degree of transmission of the reference image Y is high or low.

[0091] (3) In the above-described embodiments, the keyboard instrument **10** is an electronic instrument capable of outputting the performance data P. The keyboard instrument **10** may, however, be a natural musical instrument that does not output the performance data P. In this case, a microphone collects sound emitted from the keyboard instrument **10**, which is a natural musical instrument. In this manner, the sound signal A is generated. The performance analysis circuit **45** uses the sound signal A to analyze the performance of the user U playing the keyboard instrument **10**. For example, the performance analysis circuit **45** analyzes the sound signal A to generate a time order of the performance data P. Then, the performance analysis circuit **45** compares the time order of the performance data P with the musical piece data M of the target musical piece to identify the performance point Q1. Similarly, the performance speed Q2 and the operation position Q3 are identified by analyzing the sound signal A.

[0092] (4) In the above-described embodiments, the performance analysis circuit **45** uses the performance data P to analyze the performance of the user U. The performance analysis circuit **45**, however, may use the observation image X to analyze the performance of the user U. For example, the performance analysis circuit **45** analyzes the observation image X to identify the operation position Q3. Specifically, the performance analysis circuit **45** detects the hands and fingers Hx of the user U from the observation image X, and identifies the operation position Q3 based on the positions of the hands and fingers Hx relative to the reference position R.

[0093] (5) In the above-described embodiments, the image pick-up device **21** and the display device **22** are mounted on the head of the user U. It is not essential, however, that the image pick-up device **21** and the display device **22** be mounted on the head of the user U. For example, one or both of the image pick-up device **21** and the display device **22** may be arranged in the vicinity of the keyboard instrument **10**. It is to be noted, however, that in the configuration in which the image pick-up device **21** and the display device **22** are mounted on the head of the user U, the positions and the angles of the image pick-up device **21** and the display device **22** are variable based on the

position and the angle of the head of the user U. That is, the observation image X and the reference image Y displayed on the display device **22** are variable based on the movement of the head of the user U. This configuration enables the user U to play the keyboard instrument **10** with a sense of the reference performer's existence in the real space.

[0094] (6) In the above-described embodiments, the hands and fingers Hy of the reference performer (the trainer T) are displayed on the display device **22**. It is also possible, however, that the upper half of the body or the whole body of the reference performer is displayed on the display device **22** as the reference image Y. Also in the above-described embodiments, the reference image Y includes an image of the hands and fingers Hy of the reference performer and an image of the keyboard By. An image of the keyboard By, however, may be omitted from the reference image Y. [0095] (7) In the above-described embodiments, the user U and the reference performer play a common target musical piece. The user U and the reference performer may, however, play different musical content. For example, when a target musical piece has a plurality of performance parts, the user U may play a first part and the reference performer may play a second part. Each of the first part and the second part is one or more performance parts of a target musical piece.

[0096] (8) The information processing system **30** may be implemented by, for example, a server apparatus that communicates with a terminal device such as a smartphone and a tablet terminal. The image pick-up device **21** is incorporated in or connected to the terminal device. The terminal device sends the observation image data Dx and the image pick-up device **21** to the information processing system **30**. The observation image data Dx is generated by the image pick-up device **21**, and the performance data P is generated by the keyboard instrument **10**. The information processing system **30** generates the guide image data Dz based on the observation image data Dx and the performance data P, similarly to the information processing system **30** according to any of the above-described embodiments. Then, the information processing system **30** sends the guide image data Dz to the terminal device. The terminal device displays, on the display device **22**, a guide image Z indicated by the guide image data Dz. The display device **22** may be incorporated in, for example, the terminal device.

[0097] (9) As described above, the functions of the information processing system **30** according to any of the above-described embodiments are implemented by cooperation of a single processor or a plurality of processors constituting the control device **31** with a program stored in the storage device **32**. The program exemplified in the above-described embodiments may be provided in the form of a computer-readable recording medium and installed in a computer. An example of the recording medium is a non-transitory recording medium, and a preferable example is an optical recording medium (optical disc) such as a CD-ROM. The recording medium, however, encompasses any other recording media, such as a semiconductor recording medium and a magnetic recording medium. As used herein, the term “non-transitory recording medium” is intended to mean any recording medium that is other than a transitory, propagating signal. A volatile recording medium is encompassed within the non-transitory recording medium. The program may be distributed from a distribution device via a communication network. In this case, a recording medium that stores the program in the distribution device corresponds to the non-transitory recording medium.

H: Additional Notes

[0098] The above-described embodiments can be exemplified by the following configurations.

[0099] One form (form 1) of the present disclosure is an information processing system that includes an image obtaining circuit and a display control circuit. The image obtaining circuit is configured to obtain observation images of a first keyboard of a first keyboard instrument. The display control circuit is configured to display, on a display device, the observation images and reference images. The reference images include moving images of at least one hand and one or more fingers of a reference performer who is playing a second keyboard of a second keyboard instrument. The at least one hand and the one or more fingers of the reference performer are

displayed overlapping the first keyboard included in the observation images. With this configuration, the hands and fingers of the reference performer shown in the reference image are displayed to overlap the keyboard shown in the observation image. This configuration enables the user to play the keyboard instrument while looking at the hands and fingers of the reference performer superimposed on the keyboard of the keyboard instrument.

[0100] In a specific example (form 2) of form 1, the display device is mounted on a head of a user of the information processing system. In this form, the positions and the angles of the image pick-up device and the display device are variable based on the position and the angle of the head of the user. Specifically, the position and the angle of the keyboard is variable in the reference image displayed on the display device along with the movement of the head of the user. This configuration enables the user to play the keyboard instrument with a sense of the reference performer's existence in the real space.

[0101] In a specific example (form 3) of form 1 or 2, the reference images include the second keyboard played by the reference performer. The display control circuit is configured to display the reference images on the display device such that the second keyboard overlaps the first keyboard shown in the observation images. In this form, the keys of the reference keyboard are displaced based on a motion of the reference performer, and such reference keyboard is superimposed on the keyboard in the real space. This configuration enables the user to visually see the hands and fingers of the reference performer with a sense of the reference performer's actual operation of the keyboard.

[0102] In a specific example (form 4) of one of forms 1 to 3, the information processing system further includes a reference identification circuit analyze the observation images to identify a reference position relative to the first keyboard instrument included in the observation images. The display control circuit is configured to display the reference images at a position corresponding to the reference position. In this form, the reference image is displayed at a position corresponding to the reference position on the keyboard instrument. This configuration ensures that the hands and fingers of the reference performer are arranged at positions appropriate relative to the keyboard instrument in the real space.

[0103] In a specific example (form 5) of form 4, the reference identification circuit is further configured to detect an end portion of the first keyboard included in the observation images and determine a position of the end portion as the reference position. The (left or right) end portion of the keyboard instrument is clearly distinguished in terms of optical properties such as color from parts of the keyboard instrument which parts are located on the outer sides of the keyboard (examples of such parts are wooden clappers and wooden arm pieces). Therefore, the end portion of the keyboard is detected accurately and easily from an image of the keyboard. Thus, in this form, the position of the end portion of the keyboard is identified as a reference position. This form ensures that the reference position is identified accurately and easily.

[0104] In a specific example (form 6) of form 4, the reference identification circuit is further configured to detect a mark on the first keyboard instrument included in the observation images and determine a position of the mark as the reference position. A mark, such as a trademark, on a keyboard instrument is in many cases formed at a particular position on the keyboard instrument (for example, a center portion in the lateral direction of the housing of the keyboard instrument). In the above-described form, the position of the mark on keyboard instrument is identified as the reference position. This form ensures that the reference position is identified accurately even if the end portion of the keyboard portion can not be detected.

[0105] In a specific example (form 7) of form 4, the reference identification circuit is further configured to detect an operated key from among a plurality of keys of the first keyboard shown in the observation images, and determine a position of the operated key as the reference position. In this form, the position of the key operated by the user is identified as the reference position. This form ensures that the reference position is identified accurately even if the end portion of the

keyboard instrument or the mark on the keyboard instrument can not be detected.

[0106] In a specific example (form 8) of one of forms 1 to 7, the display control circuit is further configured to control a numerical value of a transmission parameter of the reference images. In this form, a numerical value of a transmission parameter of the reference image is controlled. This control ensures such a usage form that the hands and fingers of the reference performer are prioritized for attention from the user, and such a usage form that the user itself is prioritized for attention from the user. These usage forms are switchable to and from each other. This switchable configuration is in contrast to a configuration in which the numerical value of the transmission parameter of the reference image is fixed.

[0107] In a specific example (form 9) of form 8, the information processing system further includes a performance analysis circuit configured to determine whether the first keyboard instrument is being played. In a case where it is determined that the first keyboard instrument is not being played, the display control circuit is further configured to set the transmission parameter to a first value. In a case where it is determined that the first keyboard instrument is being played, the display control circuit is further configured to set the transmission parameter to a second value that is higher in transmission degree than the first value. In this form, while the user is playing the keyboard instrument, the transmission parameter is set to the second value, which is higher in transmission degree than the first value. This form enables the user to prioritize and focus on the hands and fingers of the user itself. This form is in contrast to a configuration in which the transmission parameter is fixed to the first value. While the user is not playing the keyboard instrument, the transmission parameter is set to the first value, which is lower in transmission degree than the second value. This form enables the user to easily look at the reference image. This form is in contrast to a configuration in which the transmission parameter is fixed to the second value.

[0108] In a specific example (form 10) of form 8, in a first motion mode, the display control circuit is further configured to set the transmission parameter to a first value. In a second motion mode, the display control circuit is further configured to set the transmission parameter to a second value that is higher in transmission degree than the first value. In this form, in the second motion mode, the transmission parameter is set to the second value, which is higher in transmission degree than the first value. This form enables the user to prioritize and focus on the hands and fingers of the user itself. This form is in contrast to a configuration in which the transmission parameter is fixed to the first value. In the first motion mode, the transmission parameter is set to the first value, which is lower in transmission degree than the second value. This form enables the user to easily look at the reference image. This form is in contrast to a configuration in which the transmission parameter is fixed to the second value.

[0109] In a specific example (form 11) of one of forms 1 to 10, the information processing system further includes a performance analysis circuit configured to identify an operation position on the first keyboard of the first keyboard instrument. The display control circuit is further configured to display the reference images in an area of the first keyboard that is higher or lower in sound than a predetermined area of the keyboard, the predetermined area including the operation position. In this form, the reference image is displayed in an area of the keyboard which area is higher or lower in sound than the operation position of the user. That is, the reference image is displayed without overlapping the hands and fingers of the user. This form enables the user to easily look at the reference image while playing the keyboard instrument.

[0110] In a specific example (form 12) of one of forms 1 to 11, the information processing system further includes a performance analysis circuit configured to identify a performance point in a musical piece. The display control circuit is further configured to display, on the display device, a performance-point part of the reference images that correspond to the performance point in the musical piece. In this form, the reproduction of the reference image proceeds along with the user's performance of a musical piece. This form enables the user to play the musical piece with a sense

of cooperating with the reference performer on the keyboard instrument.

[0111] In a specific example (form 13) of form 12, the performance analysis circuit is further configured to identify a performance speed, and the display control circuit is further configured to control a reproduction speed of the reference images based on the performance speed. In this form, the reproduction of the reference image proceeds along with the user's performance of a musical piece. This form enables the user to play the musical piece with a sense of cooperating with the reference performer on the keyboard instrument.

[0112] In a specific example (form 14) of one of forms 1 to 13, the information processing system further includes a reception control circuit configured to receive, using a communication device, moving image data indicating a motion made by the reference performer. The display control circuit is configured to display, on the display device, the reference image indicated by the moving image data. In this form, while an image of the reference performer's motion of a musical performance is being picked up, a reference image showing this motion can be displayed on the user's display device. This form enables the user to play the keyboard instrument with a sense of the reference performer's existence near the user, even though the reference performer is actually remote from the user.

[0113] Another form of the present disclosure is a computer system implemented method of processing information. The method includes obtaining observation images of a first keyboard of a first keyboard instrument. The method also includes displaying, on a display device, the observation images and reference images, the reference images including a moving image of at least one hand and one or more fingers of a reference performer who is playing a second keyboard of a second keyboard instrument, such that the at least one hand and the one or more fingers of the reference performer are displayed overlapping the first keyboard included in the observation images.

[0114] Another form of the present disclosure is a non-transitory computer-readable recording medium storing a program that, when executed by a computer system, causes the computer system to perform a method including obtaining observation images of a first keyboard of a first keyboard instrument. The method also includes displaying, on a display device, the observation images and reference images, the reference images including a moving image of at least one hand and one or more fingers of a reference performer who is playing a second keyboard of a second keyboard instrument, such that the at least one hand and the one or more fingers of the reference performer are displayed overlapping the first keyboard included in the observation images.

Claims

1. A computer executable method comprising: obtaining observation images of a first keyboard of a first keyboard instrument; and displaying, on a display device, the observation images and reference images, the reference images including moving images of at least one hand and one or more fingers of a reference performer who is playing a second keyboard of a second keyboard instrument, wherein the at least one hand and the one or more fingers of the reference performer are displayed overlapping the first keyboard included in the observation images.
2. The method according to claim 1, wherein the reference images show a fingering model for a musical piece.
3. The method according to claim 1, wherein the reference images show how the at least one hand and the one or more fingers of the reference performer are playing a musical piece.
4. The method according to claim 1, further comprising: identifying a performance point in a musical piece; and displaying, on the display device, a performance-point part of the reference images that correspond to the performance point in the musical piece.
5. The method according to claim 1, further comprising: identifying a performance speed of a musical piece; and controlling a reproduction speed of the reference images based on the

performance speed of the musical piece.

6. The method according to claim 1, further comprising: generating guide images by synthesizing each observation image of the observation images with a corresponding reference image of the reference images; and displaying, as the observation images and the reference images, the generated guide images.

7. The method according to claim 1, further comprising adjusting a size or an angle of the least one hand or at least one finger of the one or more fingers of the reference performer based on a size or an angle of the first keyboard.

8. The method according to claim 1, further comprising controlling a numerical value of a transmission parameter, corresponding to a degree of transparency, of the reference images.

9. The method according to claim 8, further comprising: determining whether the first keyboard instrument is being played; in a case where it is determined that the first keyboard instrument is not being played, setting the numerical value of the transmission parameter to a first value; and in a case where it is determined that the first keyboard instrument is being played, setting the numerical value of the transmission parameter to a second value that is higher in transmission degree than the first value.

10. The method according to claim 1, wherein the reference images including moving images of the at least one hand and the one or more fingers of the reference performer and the second keyboard of the second keyboard instrument.

11. The method according to claim 1, wherein the reference images including moving images of the reference performer.

12. A system comprising: a memory configured to store a program; and a processor communicatively connected to the memory and configured to execute the stored program to: obtain observation images of a first keyboard of a first keyboard instrument; and display, on a display device, the observation images and reference images, the reference images including a moving image of at least one hand and one or more fingers of a reference performer who is playing a second keyboard of a second keyboard instrument, such that the at least one hand and the one or more fingers of the reference performer are displayed overlapping the first keyboard included in the observation images.

13. A non-transitory computer readable storage medium configured to store a program that, when executed, performs a method comprising: obtaining observation images of a first keyboard of a first keyboard instrument; and displaying, on a display device, the observation images and reference images, the reference images including moving images of at least one hand and one or more fingers of a reference performer who is playing a second keyboard of a second keyboard instrument, wherein the at least one hand and the one or more fingers of the reference performer are displayed overlapping the first keyboard included in the observation images.
